

# CE 310 — Capstone Report

Team submission | Due: Saturday, December 6 by midnight — submit to D2L

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## Overview

Your individual lab notebook (the capstone lab notebook) is where you ran the analysis. This report is different: it's where your team steps back from the code and writes the deliverable your client actually asked for — a short, professional document that answers their specific questions in plain language, backed by your numbers.

All three members of your team analyzed the same team-chosen dataset independently. Before writing this report, compare your individual results (Model 2 coefficients, t-test output, Monte Carlo results). Small differences from rounding are normal; if your numbers disagree substantially, find out why before you write — the report should present one team-agreed set of findings, not three conflicting ones.

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## How to Submit

☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

| Requirement | Detail                                                                                                   |
|-------------|----------------------------------------------------------------------------------------------------------|
| Format      | PDF, one file per team                                                                                   |
| Length      | 2–3 pages (excluding an optional appendix)                                                               |
| Submission  | One team member submits on behalf of the whole team. List all three full names at the top of the report. |
| File name   | lastname1_lastname2_lastname3_capstone_report.pdf (alphabetical by last name)                            |
| Deadline    | Sat Dec 6, midnight — D2L Dropbox                                                                        |

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## Required Content

Your own client brief (your client brief) lists the 2–3 specific questions your team wrote for your stakeholder. Your report must answer every one of them explicitly — not just present your analysis and hope the client can connect the dots.

1. Executive Summary (3–5 sentences) The bottom line, up front: what did you find, and what should the client do? A busy stakeholder should be able to read only this paragraph and walk away with the key takeaway.
  2. Dataset & Client Overview (short paragraph) What the dataset is and where it's from, who the client is, dataset size (ANSWER\_A1 rows), and the mean of your outcome variable (ANSWER\_A3) in plain terms.
  3. Findings — answer each client question directly For each numbered question in your client brief, write a short subsection: state the question, then answer it using your team's regression (Model 1 vs. Model 2), hypothesis test, and uncertainty/Monte Carlo results. Include the specific numbers (coefficients, p-value,  $R^2$ , P10/P90 range) — a client-facing report still needs to show its work, just without code or statistical jargon left unexplained.
  4. Recommendation One concrete, quantified, actionable recommendation that follows from your findings (e.g., a specific operational change and its estimated savings, or a specific risk to budget for given the P10/P90 range). Avoid generic advice (“reduce energy use”) — tie it to a number from your analysis.
  5. Appendix (optional, doesn't count toward the 2–3 page limit) A table of key statistics (Model 1/Model 2 coefficients,  $R^2$ , t-statistic, p-value, cost P10/P50/P90) for reference, if you want to keep the main body number-light.
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## What This Is Not

- Not a copy-paste of your notebook's Section E memo — that was individual and brief (3 paragraphs); this is a team-synthesized, client-facing document.
  - Not a place for new analysis — use the numbers your team already produced in the lab notebook.
  - Not graded on length — a tight, well-organized 2 pages beats a padded 3.
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## Grading

See the capstone report rubric. Worth 40 raw points, scaled to the Capstone Report category in the course gradebook — separate from your individual notebook grade (40 pts) and your team's presentation grade (60 pts, Week 16).